

Make It Do Something

Week 6 · General AI Fluency · Plain-words explainer

What I built

I added exactly one dynamic feature to my portfolio: a working contact form. A visitor can enter their name, email address, and message and submit it from the live portfolio. The message is sent to my inbox instead of only changing something on the page.

What a backend is

The backend is the part of a web application that runs away from the visitor's browser. It can receive data, check it, keep secrets, talk to other services, and return a result. In this feature, the backend is a small Next.js server route. The browser should not contain the email-service API key, so the server uses that secret instead.

What the feature does

The form collects three pieces of information: name, email, and message. When the visitor presses **Send message**, the browser sends those values to my Next.js **/api/contact** route. The route validates the basic input and then sends the message through the Resend Email API. The recipient is my portfolio inbox, and the visitor's email is used as the reply-to address so I can respond to them.

How the data flows

1	Visitor	Types name, email, and message into the contact form.
2	Browser	Sends the form data to /api/contact with a POST request.
3	Next.js backend	Checks the input and uses the private RESEND_API_KEY.
4	Resend	Receives the server request and delivers the email.
5	My inbox	Receives the submitted message; replies go to the visitor's email.

Why there is no database

I did not add a database because this feature does not need one. The assignment asks for one real dynamic feature, not a full application. The message only needs to travel from the form to my inbox, so adding storage, authentication, or other backend systems would add complexity without helping the feature.

What I understand from the build

The important part is that the browser is not sending the email directly with a secret key. The browser sends the message to my own server route. The server can safely use the private API key to call the email service. That separation is the main reason the backend is useful here.

Verification note

The code is deployed from the portfolio repository, but the final end-to-end test depends on adding the private RESEND_API_KEY to the Vercel project. The final submission should only claim the feature as working after a real form submission has arrived in the inbox.

Feature scope: exactly one — working contact form.